Amish Gogia

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Education

Eidgenössische Technische Hochschule Zürich (ETH Zürich) — *Zürich, Switzerland*
Master of Science in Physics — *2025 - Present*

- **Research Interests:** Experimental Particle Physics and Machine Learning.

Delhi Technological University (DTU) — *New Delhi, India*
Bachelor of Technology in Engineering Physics — *2021 - 2025*

- **CGPA: 9.16/10**, graduated top 4% of institute in a batch of ~2000 people.
- **Minor in Machine Learning**.
- **Selected Coursework:** Classical and Quantum Mechanics, Mathematical Physics, Computational Methods, Quantum Information Theory, Cosmology and Astrophysics, Computer Vision, Neural Networks and Deep Learning.

Research Experiences

High Energy Physics Laboratory, EPFL — *Lausanne, Switzerland*
Swiss Summer Student Particle Physics Program (S3P3) — *July 2026 – September 2026*
Advisors: *Prof. Dr. Laurent Dufour and Dr. Andrea Villa*

- Studied CP violation in the B^0 and B_s^0 systems with FCC-ee simulation samples.
- Worked on designing the analysis workflow from scratch using the FCCAnalyses framework.
- Calculated the precision on a_{sl}^s which can be achieved with the FCC-ee and compared that to current world average values.
- Awarded **Best Talk** at the S3P3 final event and **Best Poster** at the Summer Poster Session for Interns in Physics at EPFL.

Institute for Particle Physics and Astrophysics, ETH Zürich — *Zürich, Switzerland*
Semester Project — *Feb 2026 – September 2026*
Advisors: *Prof. Dr. Thea Klaeboe Aarrestad and Patrick Odagiu*

- Worked on symmetry-aware jet tagging models to probe improved performance for the CMS detector.
- Imposed $\mathrm{SO}(2)$, $\mathrm{SO}(3)$ and Lorentz invariance on DeepSets, MLP and Transformer through data augmentation and regularisation loss, and compared their performance.
- Designed the full ML Ops pipeline using PyTorch Lightning and HYDRA for efficient development.

High Energy Physics Software Foundation — *Remote*
HSF-India Fellow — *July 2025 – Jan 2026*
Advisor: *Dr. Bhawna Gomber (University of Hyderabad)*

- Contributed to development of analysis pipeline for Run 3 analysis of $h \to b\bar{b}$ to look for dark matter signatures.
- Gained experience in using the Coffea framework to analyse NanoAOD files and also in running samples using HTCondor.

Department of Applied Physics, DTU — *Delhi, India*
Research Assistant — *Dec 2024 – May 2025*
Advisor: *Dr. Deshraj Meena*

- Applied machine learning to predict gas-sensing performance of doped SnO_2 nanostructures.
- Trained five models (RF, CatBoost, NN, XGBoost, Stacking) using literature-derived data.
- Assessed model accuracy using RMSE and R^2 metrics.
- Used trained models to study the effects of dopant level, temperature, and gas concentration.

Department of Applied Mathematics and Applied Physics, DTU *Delhi, India*
Bachelor Thesis *Sept 2024 – May 2025*
Advisors: *Dr. Satyabrata Adhikari and Dr. Mukhtiyar Singh*

- Refined the realignment (CCN) criterion to obtain a necessary and sufficient entanglement condition for two-qubit X-states.
- Derived a modified realignment inequality using singular values and PPT eigenvalue structure.
- Expressed the new criterion directly in terms of X-state density matrix elements.
- Demonstrated improved entanglement detection on a generalized X-state example compared to PPT and standard realignment criteria.
- Established parameter regions where the modified criterion detects states missed by the standard method.

High Energy Physics Laboratory, EPFL *Lausanne, Switzerland*
Project Student *June 2024 – July 2024*
Advisors: *Prof. Dr. Lesya Shchutska and Dr. Yunxuan Song*

- Searched for exotic hadrons in B-meson decays using processed LHCb Run 1 and Run 2 datasets.
- Trained Boosted Decision Trees with TMVA for optimized signal–background separation.
- Performed 1D and 2D fits in RooFit to extract signal yields.
- Performed a Partial Wave Analysis using the TF-PWA framework.
- Actively contributed to weekly discussions with supervisors and lab members.

Research Papers

Realignment Criterion: A Necessary and Sufficient Condition for Two-Qubit X-States *Preprint*
Amish Gogia*, Aakhyat Bagga, Satyabrata Adhikari, Mukhtiyar Singh* *2025*

Machine Learning Driven Response Prediction of Doped SnO_2-Based Chemiresistive Gas Sensors Towards Detection of Multiple Gases *Preprint*
Amish Gogia*, Karthik Venkateshwaran, Anubhav Bhoumik, Ayan Sajwan, Mohit Kumar, Deshraj Meena* *2025*

Awards and Scholarships

Best Talk Award *2026*
Swiss Summer Student Particle Physics Program (S3P3) final event
Best Poster Award *2026*
Summer Poster Session for Interns in Physics at EPFL
Working Internships in Science and Engineering (WISE) Scholarship *2024*
German Academic Exchange Service (DAAD)
Summer Research Fellowship Program *2024*
Indian Academy Of Sciences, Bengaluru
Merit Scholarship Award *2022–2023*
Delhi Technological University – Awarded for achieving the top academic rank among 120 students.

Skills

HEP software: ROOT/PyROOT, Coffea, uproot, awkward-array, RooFit, TMVA, FCCAnalyses, HTCondor.
Machine learning: PyTorch, PyTorch Lightning, Hydra, TensorFlow, scikit-learn.
Coding: Python, C++, Bash, SQL, Slurm.
Languages: English (full professional proficiency), German (A2), Hindi (native/bilingual proficiency).